LINUX IN THE OPEN SOURCE ECOSYSTEM



by Abhiram B K



What is Open Source?

Open source refers to software where the original source code is made freely available, allowing anyone to inspect, modify, and redistribute it. This transparency fosters global collaboration, leading to continuous improvement and innovation.

Think of it like a community garden: everyone can contribute to planting, tending, and sharing the harvest, benefiting the whole community.

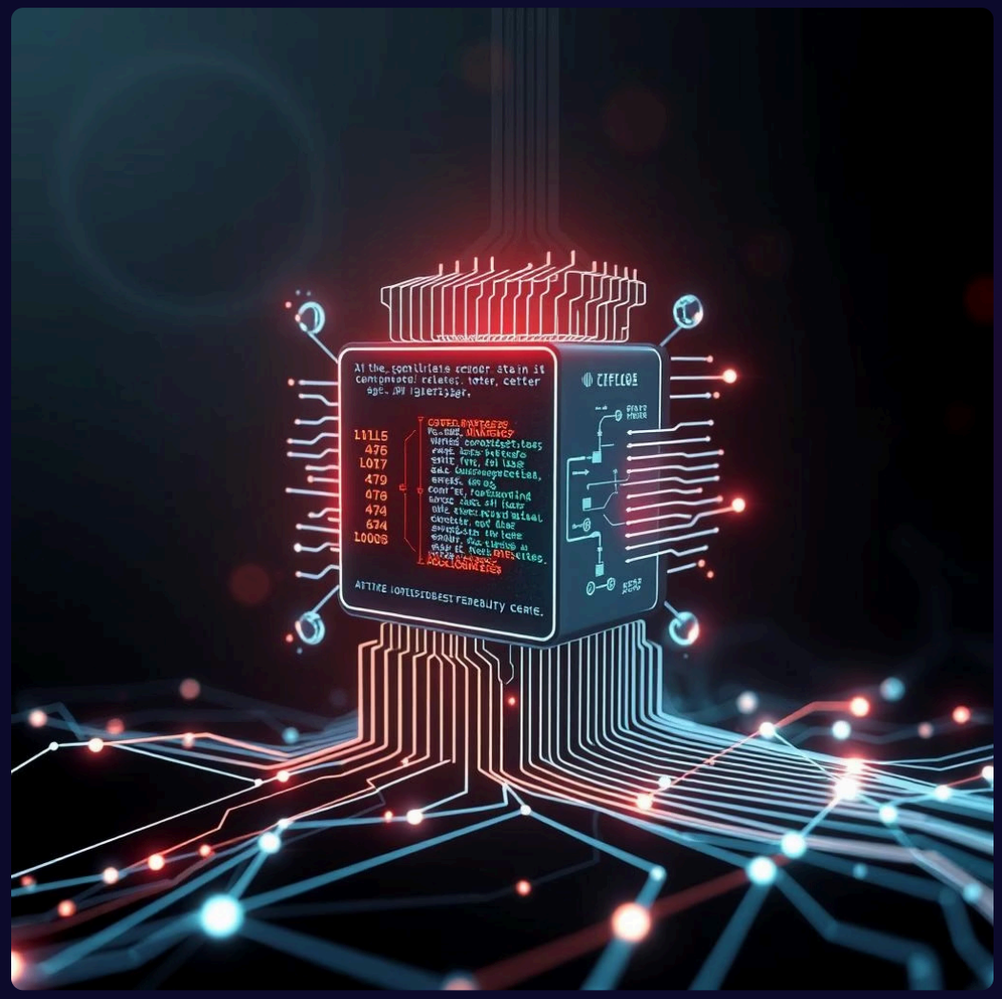
Instead of a locked-down product you can’t see inside, you get:

- The source code
- The rights to study it
- The rights to share it
- The rights to change it (or break it, no judgment)



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What is Linux? More Than Just an Operating System



Powering 90% of global cloud workloads, Linux is the unseen force behind modern technology, from Android to supercomputers and IoT devices, built on foundational open-source principles.

A Kernel: Core of an OS

The essential part of an operating system, managing hardware. The **Linux we use daily** isn't just the kernel; it's the kernel plus a massive collection of open-source tools (shells, compilers, file utilities). This is why you sometimes hear "GNU/Linux."

Free and Open Source (FOSS)

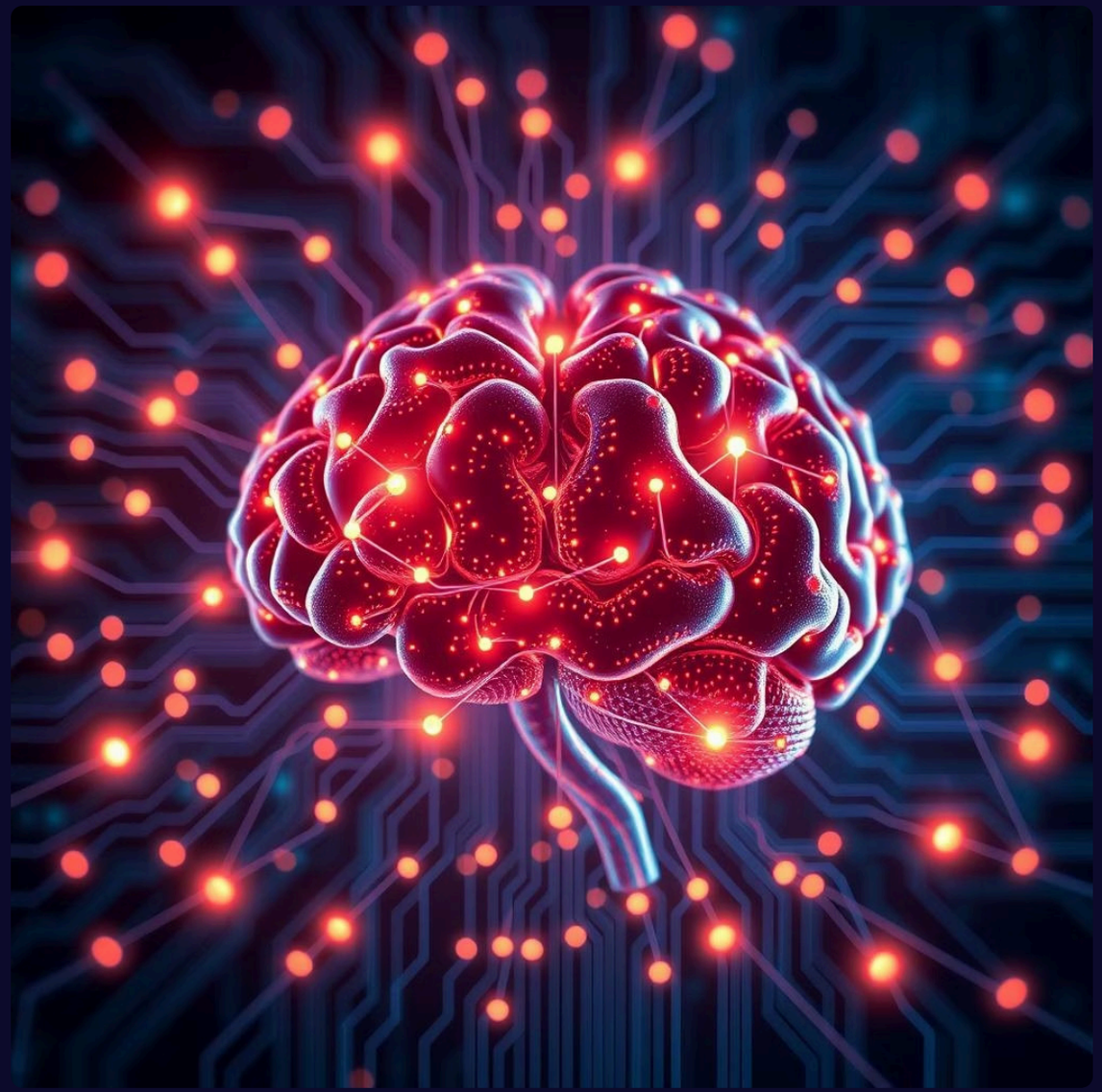
Code is freely available, modifiable, and shareable.

Distributed as "Distributions"

Complete OS packages like Ubuntu and Fedora.

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What is a Kernel?



The kernel is like the heart (or brain?) of the operating system.

It handles:

- Talking to hardware
- Managing memory
- Scheduling processes
- Enforcing Security

  (sorry Hackers 😁)

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The Fuel: The GNU Project and Free Software Ideals

Founded by Richard Stallman (1983)

Pioneering the free software movement.

Goal: Complete Free Unix-like OS

A vision for software freedom and user control.

GNU General Public License (GPL)

Ensuring software remains free and open.

Provided Essential Tools

Compilers, shells, and other core utilities.

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The Spark: Linus Torvalds and the Kernel's Birth

In 1991, Linus Torvalds, a Finnish student, began a personal project to create a free operating system kernel, inspired by MINIX. He announced his work on Usenet, inviting collaboration.



Linux Born in 1991

Started as Torvald's personal academic endeavor.

Inspired by MINIX

A small Unix-like OS for educational purposes.

Initial Code: 10,239 lines

A modest start for a massive project.

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The Synergy: GNU/Linux - A Match Made in Open Source

The Linux kernel completed the GNU Project's vision, creating a fully functional free operating system. Early "distributions" like SLS and Debian emerged, fostering a unique model of collaborative development.

1

Linux Kernel

The heart of the system.

2

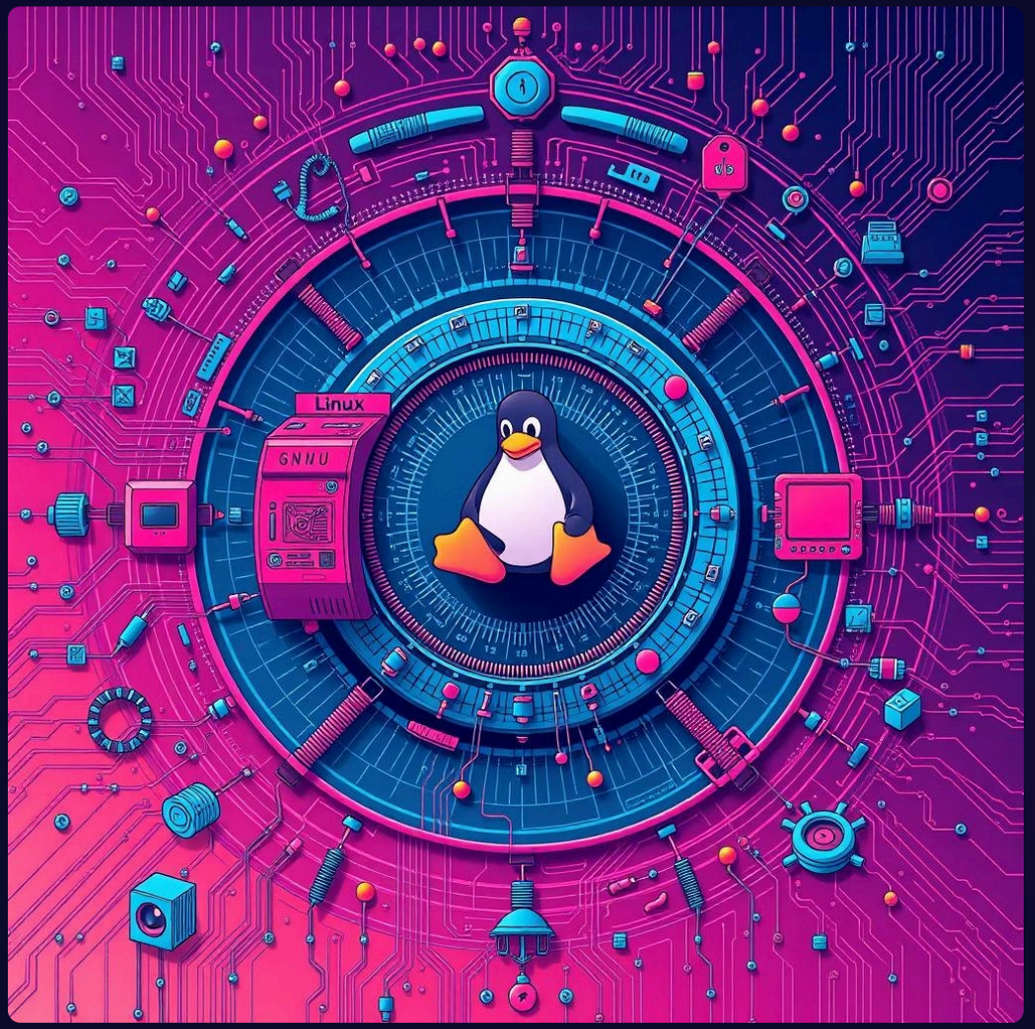
GNU Tools

The essential utilities.

3

Community

Millions of lines of code.



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From Niche to Ubiquity: Linux Goes Mainstream



Enterprise Servers

90% of cloud workloads run on Linux.



Mobile Devices

Android, powered by Linux, dominates 70%+ of smartphones.



Supercomputing

100% of the Top 500 supercomputers use Linux.



IoT & Embedded

Powers smart TVs, routers, and countless IoT devices.

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Cornerstone of Open Source: Driving Collaboration & Innovation



27.8M+

Lines of Code (2023)

The vast scale of the Linux kernel.

20k+

Developers

A global community contributing to its growth.

100%

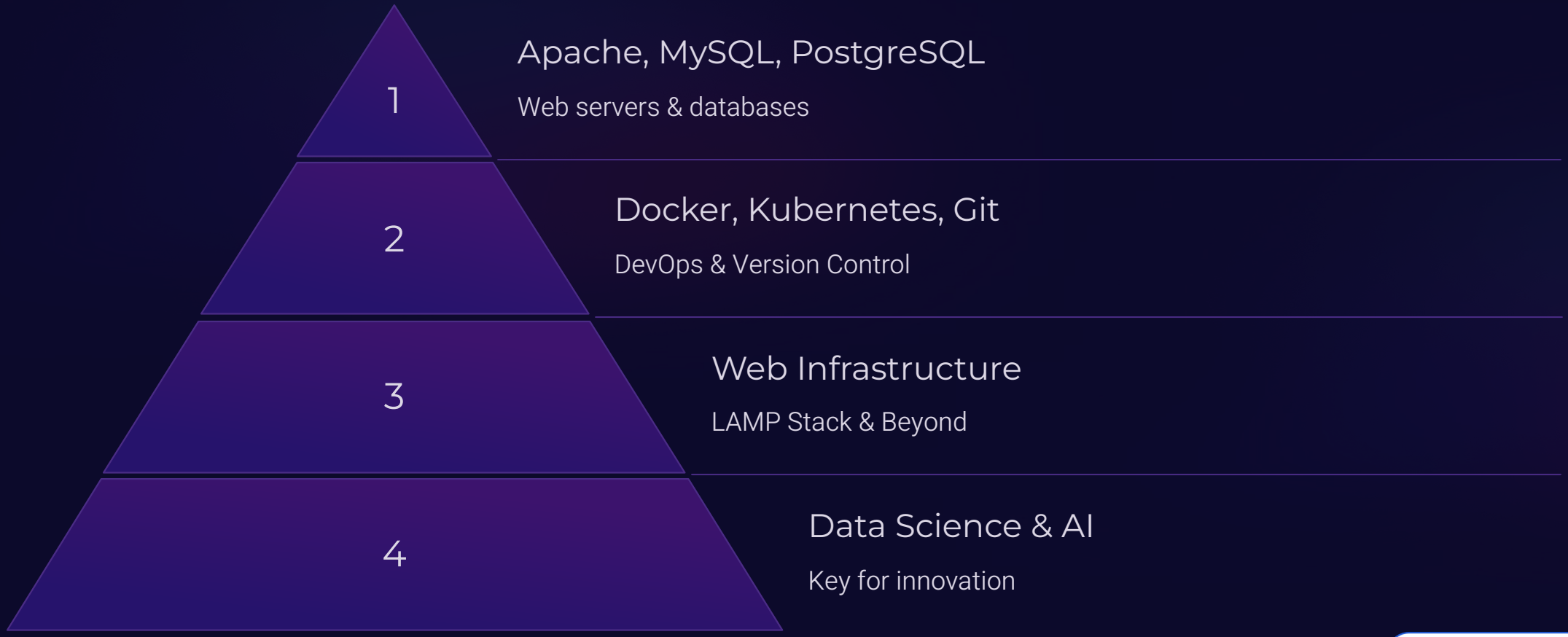
Transparent Development

A model for open collaboration and rapid iteration.

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The Ecosystem Effect: Spawning a Universe of Innovation

Linux's open architecture enabled a cascade of foundational open-source projects and technologies that reshaped modern computing.



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Beyond the Desktop: Fun Facts & Unexpected Uses

Mars Rover Perseverance

Exploring the Red Planet.

International Space Station

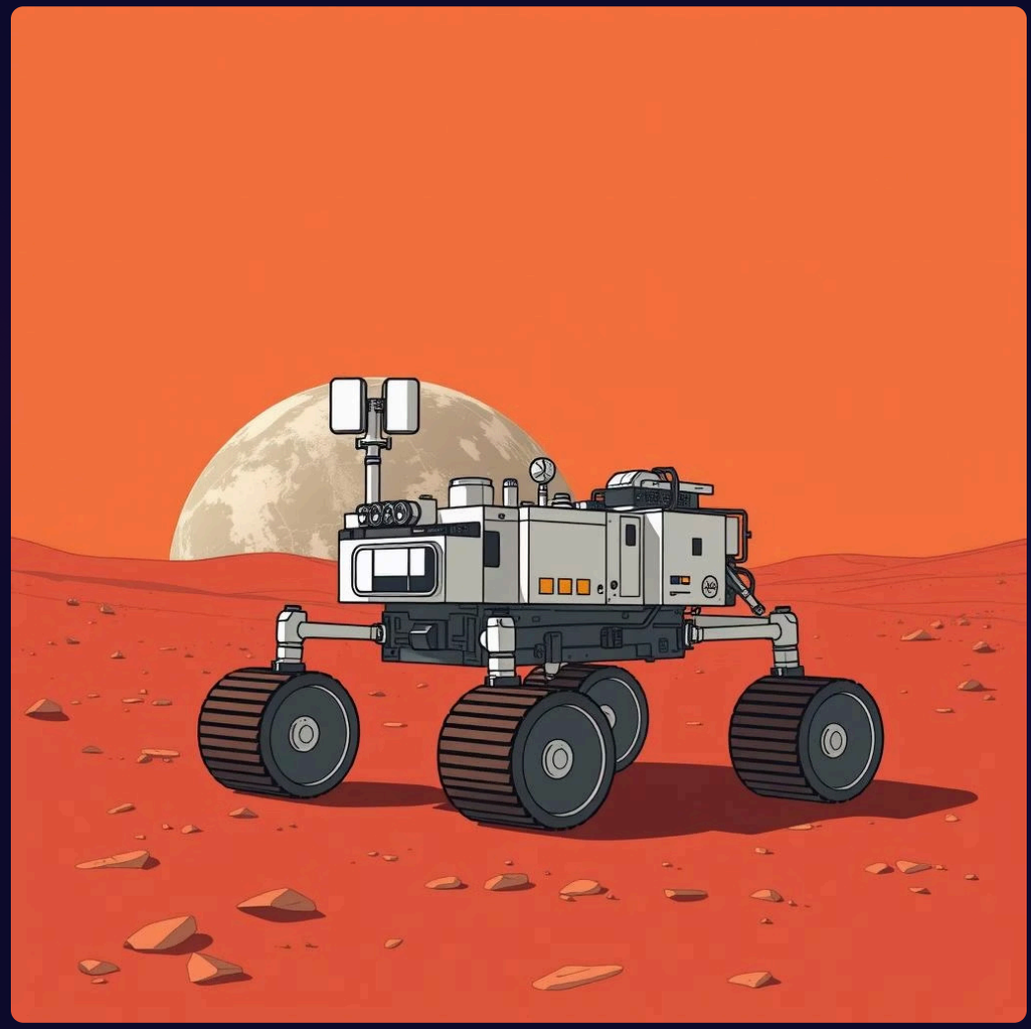
Keeping astronauts connected.

Netflix, Google, Amazon

Powering global digital giants.

ATMs & High-Speed Trains

Ensuring critical operations for the Shinkansen.



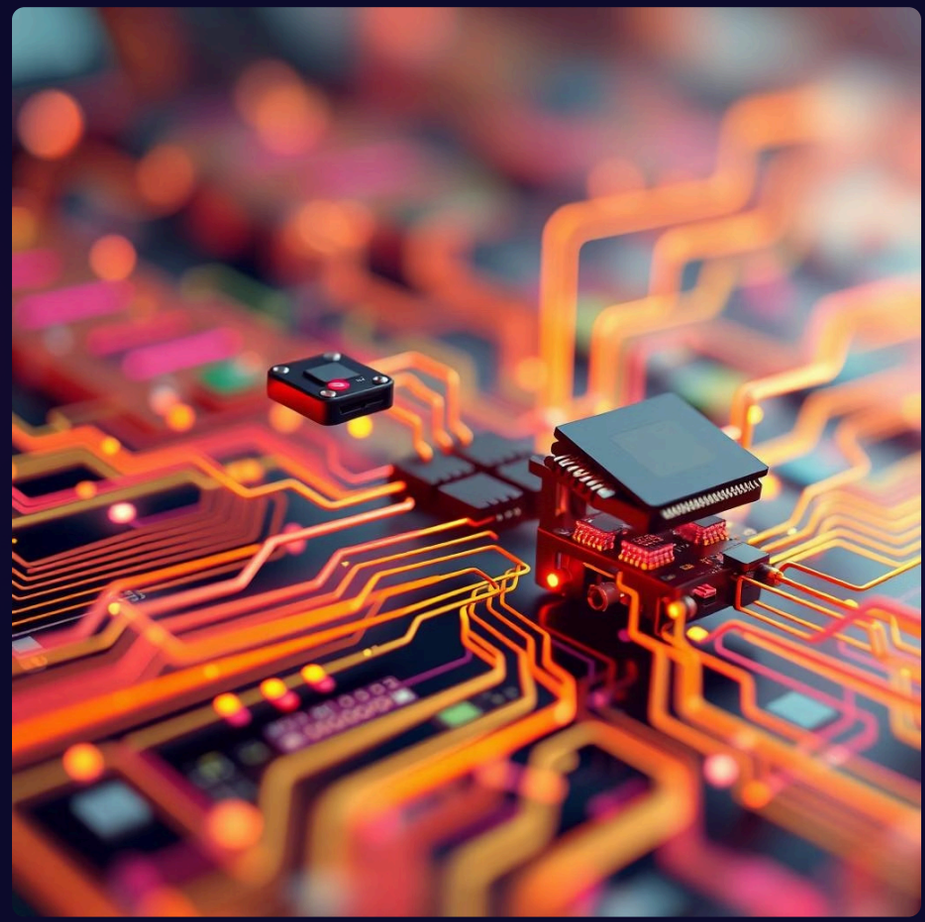
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Flexibility and Efficiency: The Power of Linux Distributions

Linux distributions offer unparalleled adaptability, allowing users to tailor their operating system precisely to their needs, whether for specialized tasks, development, or everyday computing.

The inherent lightweight nature of many Linux distributions means they can run efficiently on older hardware, revitalizing machines that would struggle with more resource-intensive operating systems.

Users can choose exactly which components and services to install, leading to lean, fast, and stable systems that excel in specific applications, from high-performance computing to embedded systems in IoT devices.



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Essential Linux Commands: Power at Your Fingertips

Navigation & Files

ls: List contents.

cd: Change directory.

mkdir: Create new folders.

rm: Delete files.

cp: Copy files.

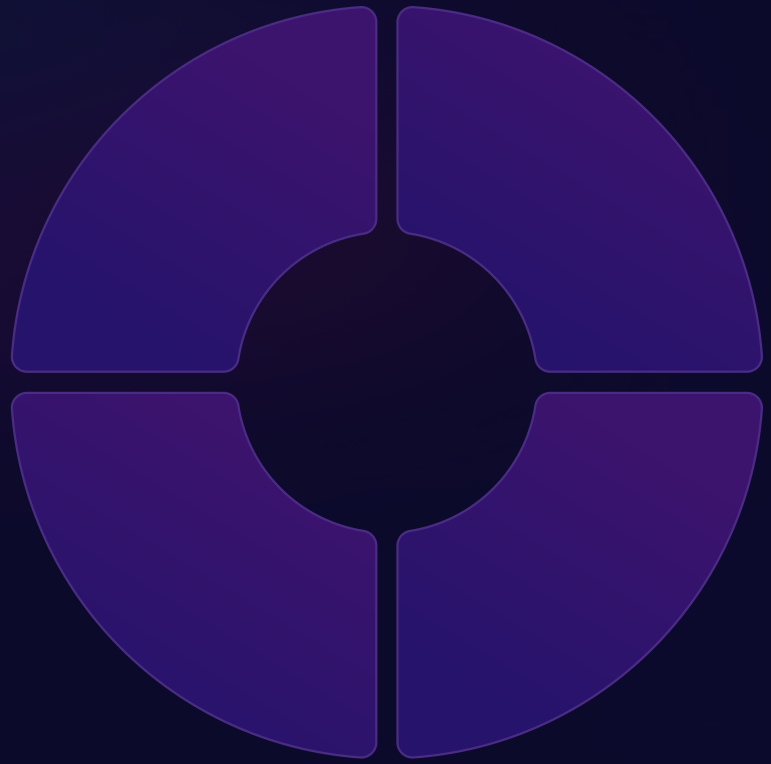
Networking & Processes

ping: Test network connectivity.

ifconfig (or **ip addr**): View network interfaces.

ps: Display running processes.

kill: Terminate processes.



System & Information

grep: Search for text.

man: Access command manuals.

df: Report file system disk space usage.

du: Estimate file space usage.

top: Display Linux processes.

Permissions & Users

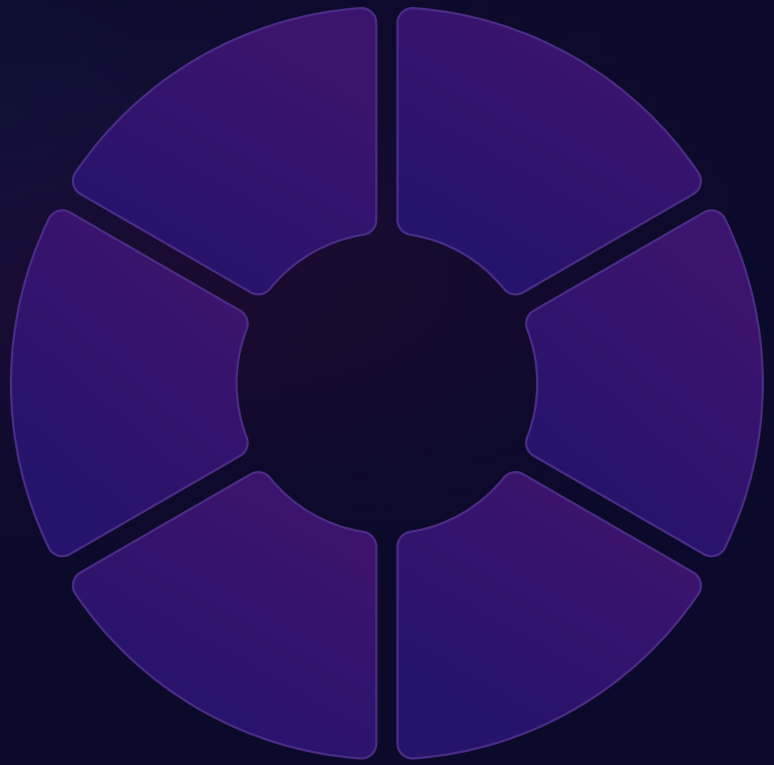
chmod: Change file permissions.

chown: Change file owner.

sudo: Execute commands as superuser.

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A Simple Demo of fun things with Linux



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Conclusion: Linux - A Testament to Open Collaboration

A Global Success Story

Driven by collective effort and shared vision.

Shaping the Future of Tech

Continual evolution and adaptation.

Power of Community Innovation

A vibrant ecosystem of contributors.

Enduring Force for Open Computing

Promoting freedom and accessibility in technology.



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References and Resources

Explore the foundational sources that shaped this presentation and delve deeper into the world of Linux and open source.

Key Resources:

Linux Kernel Documentation

GNU Project Official Website

Open Source Initiative

The Linux Foundation

r/linux Subreddit

The Linux Command Line Book (Free)

Ubuntu Manpages



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Thank You for Your Time and Engagement!

I sincerely appreciate your presence and engagement throughout this presentation on Linux and the open-source ecosystem.

Your interest fuels my shared journey towards collaborative innovation.



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